

The genome of the Pacific acorn barnacle provides insights into the evolution of extremely large populations

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Abstract

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Introduction

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Results

Genome assembly

We assembled a chromosome-level reference genome for the Pacific acorn barnacle using PacBio Hi-Fi long reads and Hi-C chromosome conformation capture. Our initial contig-level assembly, generated using `hifiasm` [Cheng2021; Cheng2022] with default parameters, generated sequences with a total length of 1.64 Gbp, composed of 4,212 contigs, and a contig N50 of 734 kbp. We also assessed gene-completeness using BUSCOs [Manni2021; Tegenfeldt2024], as implemented in `compleseam` [Huang&Li2023], resulting in a completeness of 95.66% (S:45.31%; D:50.35%; F:0.79%; M:3.55%; N:1,013) for the `arthropoda_odb10` single-copy ortholog set (Table 2). The statistics for this initial assembly, with twice the expected genome size and over 50% duplicated BUSCOs, suggested the presence of a large fraction of haplotig sequences, reflecting the high heterozygosity expected for the species.

Following optimization, both in the integration of additional data types and in the increased stringency of the purging of haplotig and contaminant sequences (see Methods), we generated a contig-level assembly composed of 1,342 sequences, with a total length of 1.05 Gbp, a contig N50 of 1.35 Mbp, and a BUSCO completeness of 93.88% (S:89.04%; D:4.84%; F:1.18%; M:4.94%; N:1,013, `arthropoda_odb10`). The contig-level assembly was then scaffolded with Hi-C data using the `YAHS` software [Zhou2022]. Our scaffolding process generated an assembly with a total length of 1.05 Gbp, composed of 650 scaffolds, a scaffold N50 of 50.96 Mbp, and a BUSCO completeness of 93.98% (S:89.24%; D:4.74%; F:0.89%; M:5.13%; N:1,013, `arthropoda_odb10`).

This scaffold-level assembly was further processed by performing manual curation of the Hi-C contact map and self-correction using the software **Inspector** [Chen2021]. This curated assembly was composed of 592 fragments, a total sequence length of 1.04 Gbp, a scaffold N50 of 51.41 Mbp, and a per-base quality value (QV) of 30.49. When assessing completeness, the curated assembly was 94.08% (S:89.44%; D:4.64%; F:0.89%; M:5.03%; N:1,013) and 92.77% (S:87.77%; D:5.40%; F:2.93%; M:4.30%; N:1,536) gene-complete for the **arthropoda_odb10** and **crustacea_odb12** BUSCO datasets, respectively. Of the 1.04 Gbp of total sequence, 906.6 Mbp (86.74%) was contained in 58 scaffolds larger than 1 Mbp. Moreover, the 16 largest sequences (larger than 8.5 Mbp) contain 841.2 Mbp (80.61%) of the total sequence, and likely represent 16 putative chromosomes. This karyotype number, N=16, is in agreement with that observed for the genus *Balanus* [Austin1958] and other acorn barnacle species in the family Balanidae [Han2024].

Genome annotation

Once the assembly and scaffolding was completed, we annotated and masked repeat sequences in the *B. glandula* reference assembly using the **EarlGrey** software [Baril2024]. This analysis identified 698.6 Mbp (66.94%) of the assembly as repetitive elements (Table 3); however, the majority of these, 404.5 Mbp (38.76% of the assembly), remain unclassified among the major repeat element types. This large proportion of unclassified repeat elements is likely the product of the very small number of representative barnacle sequences among curated repeat reference libraries, reflecting barnacle’s status as a genomically understudied group. Among the classified repeats, DNA and LINE elements were the two most abundant repeats classes, comprising 8.45% and 9.16% of the total assembly, respectively.

Following repeat annotation and masking of the reference genome, we annotated protein-coding genes using short-read RNAseq and long-read PacBio Iso-Seq sequencing, combining and curating annotations generated across the two data types. This process resulted in 20,234 protein-coding genes annotated in the genome. The resulting annotation was 93.58% (S:79.76%; D:13.82%; F:1.78%; M:4.64%; N:1,013) and 84.77% (S:72.66%; D:12.11%; F:4.10%; M:11.13%; N:1,536) gene-complete when compared against the **arthropoda_odb10** and **crustacea_odb12** BUSCO datasets, respectively. Additionally, we compared our annotation against the proteome of four other barnacle species (see Methods) using **OrthoFinder** [Emms&Kelly2015; Emms&Kelly2019], resulting in the assignment of 16,978 (83.91%) *B. glandula* genes to orthologous groups across barnacles and the identification of 2,576 single-copy orthologs across all five species.

In addition to protein-coding genes, we also annotated non-coding elements in the *B. glandula* reference assembly. We identified 7,532 long, non-coding RNAs (lncRNA), 1,203 transfer RNAs (tRNA), and 24 ribosomal RNA (rRNA) genes. In total, across protein-coding and non-coding transcripts, we were able to annotate 28,993 genes in the genome.

Conserved architecture across barnacle genomes

We performed a conserved synteny analysis using the software **Genespace** [Lovell2022] to identify patterns of large-scale structure across barnacle genomes. We compared our *B. glandula* genome against the chromosome-level genome assemblies of *Amphibalanus amphitrite* (Balanidae) [Han2024] and *Pollicipes pollicipes* (Pollicipedidae) [Bernot2022]. These genomes display conservation in their large-scale organization, including correspondance between 16 one-to-one orthologous chromosomes across the three species (Figure 1). This finding both shows that our scaffolding of the *B. glandula* assembly likely reflects the biological structure of the genome and is not the product of technical artifacts, but also indicates the conservation in the genome architecture of barnacle genomes across ~165 Mya of evolution [Han 2024; Bernot2022].

Diversity in the barnacle reference assembly

We calculated estimates of diversity present in the barnacle genome by genotyping our *B. glandula* reference individual. After filtering low-quality genotypes and masking regions with annotated repeats, we obtained genotypes for 314 million sites along the genome, equivalent to 34.71% of the 904.94 Mbp present in the 58 scaffolds larger than 1 Mbp (Table 1). Among these 314 million genotyped sites, 9.08 million (2.89%) were heterozygous in the reference individual, including 8.73 million SNPs and 356 thousand indels. In total,

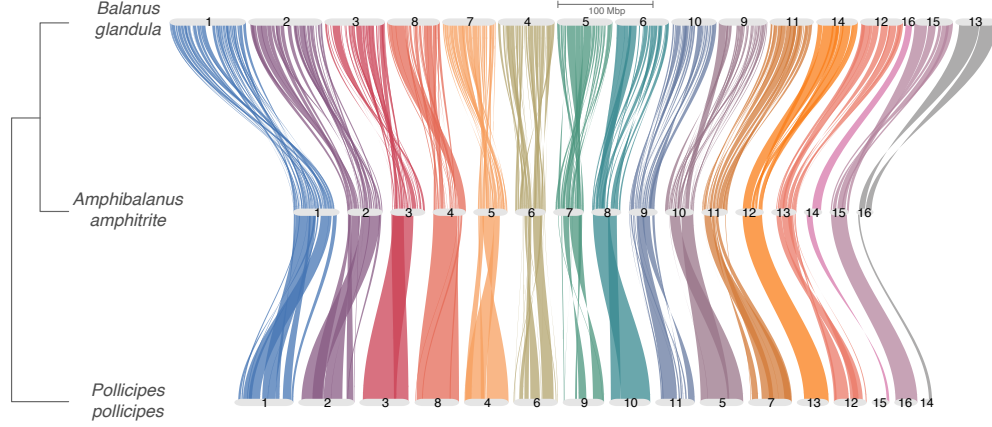


Figure 1: Conserved synteny across barnacle genomes. Genome-wide riparian synteny plots showing one-to-one correspondence across 16 orthologous chromosomes between the genomes of *Balanus glandula* (top), *Amphibalanus amphitrite* (middle), and *Pollicipes pollicipes* (bottom). Each set of colored lines shows orthologous genes across the three assemblies, colored according to their chromosome of origin. Chromosomes are drawn according their order in *A. amphitrite*. Cladogram on the left shows the evolutionary relationship among the three species.

this variation represent a mean density of 14.66 heterozygous sites per kbp (median: 6; standard deviation: 19.86). Moreover, we observed a total of 399 thousand variant sites within the span of protein-coding sequences, indicating a substantial proportion of coding variants in just a single barnacle individual.

Table 1: Genetic variation in the *B. glandula* reference individual.

Metric	Value
Reference sequence length (bp)	904,940,360
Genotyped sites post-filtering (bp)	314,105,136
Total variant sites (bp)	9,081,889
Percent variant sites (%)	2.89
SNPs (N)	8,725,877
Indels (N)	356,012
Mean variant sites per kbp (N)	14.66
Median variant sites per kbp (N)	6
St Dev variant sites per kbp (N)	19.86
Variant sites in CDSs (N)	398,773
Percent variants in CDS (%)	4.39
Mean variants per CDS (N)	4.34
Median variants per CDS (N)	1
St Dev variants per CDS (N)	11.75

Conservation of functional elements in barnacle genomes

We used the **phastCons** software [Siepel2005] on the **cactus** whole-genome alignment [Armstrong2020] between six barnacle species to identify highly conserved sites in the Pacific acorn barnacle genome. Across the 16 putative chromosomes in the *B. glandula* assembly, **phastCons** identified 196,141 highly-conserved elements (HCEs), spanning 50.85 Mbp of sequence and 4.92% of the assembled genome. In *B. glandula*, the HCEs had a mean length of 259.25 bp (median: 42 bp, standard deviation: 990.40 bp).

Following the identification of HCEs, we compared their sequence composition against the genome-wide

background (Figure 2). For this, we first overlaid the genomic coordinates of the HCEs against the *B. glandula* annotation, obtaining the proportion of annotated features among HCE sites (Figure 2A). Among our target features, including an “other” and “multiple” class for the unannotated fraction of the genome and annotation overlaps, respectively, all but one is present among HCEs. We do not observe any rRNA sites among our conserved sites; however, this is likely the product of the whole-genome alignment (given rRNA’s repetitive nature and alignment difficulty [e.g., Gillispie2004]) and not an accurate measure of conservation. Aside from the absence of rRNAs, the sequence composition of HCE largely follows the composition of the whole genome. Most annotated features do not appear substantially enriched among HCE sites (Figure 2B, Table 4). The two clear exceptions are coding sequences (CDSs) and tRNAs. CDSs display an average two-fold enrichment among HCEs (mean: 1.96, median: 1.98, standard deviation: 0.846), reflecting the conservation of coding sequences among barnacle genomes. Similarly, tRNAs are five-fold enriched in HCEs (mean: 4.99, median: 4.10, standard deviation:), the likely product of the strong functional constraint expected for these sequences.

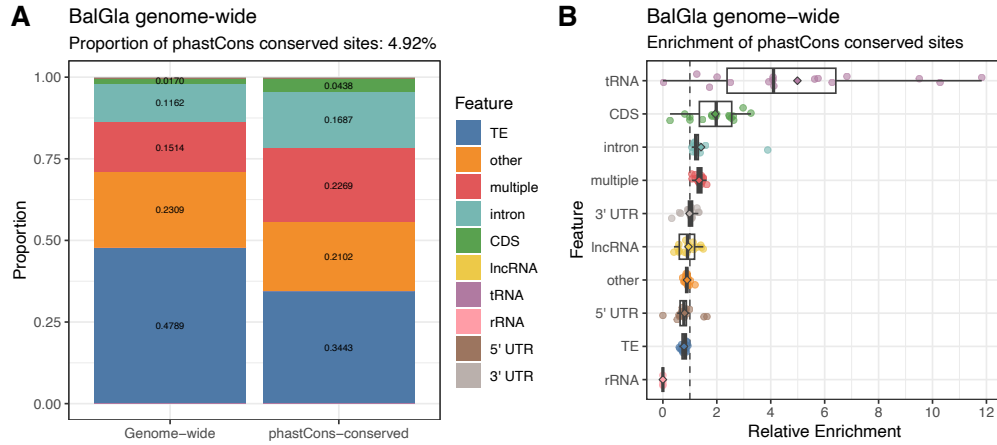


Figure 2: Highly conserved elements in the *Balanus glandula* assembly. **A.** Proportion of annotated features between the whole genome (left column) and phastCons-conserved elements (right column). Values are shown only for elements with proportions larger than 0.01. Sites that span more than one annotation type are represented by the “multiple” feature. **B.** Relative enrichment of annotated features in the highly conserved elements. The enrichment values for each individual chromosomes are shown as points. The diamond in the boxplot indicates the mean enrichment.

dN/dS

MK-test

Diversity Oregon population

Discussion

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Conclusion

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Materials and Methods

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Data Availability

https://github.com/arcolon14/balanus_genome_ms

Acknowledgments

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Supplements

Table 2: Supplemental Table: Assembly statistics for the *Balanus glandula* genome at different stages of the assembly process. BUSCO results refer to the `arthropoda_odb10` dataset.

Assembly Stage	Default contigs	Optimized contigs	Hi-C scaffolds	Curated assembly
Sequence length (Mbp)	1.64	1.05	1.05	1.04
Fragments (N)	4,212	1,342	650	592
Contig N50 (Mbp)	0.734	1.35	1.18	1.18
Scaffold N50 (Mbp)	NA	NA	50.96	51.41
BUSCO Complete (%)	95.66	93.88	93.98	94.08
BUSCO Single (%)	45.31	89.04	89.24	89.44
BUSCO Duplicates (%)	50.35	4.84	4.74	4.64
BUSCO Fragmented (%)	0.79	1.18	0.89	0.89
BUSCO Missing (%)	3.55	4.94	5.13	5.03
BUSCO size (N)	1,013	1,013	1,013	1,013

Table 3: EarlGrey high-level annotation of the *B. glandula* assembly

Classification	N Sites bp	Count	Proportion	N Distinct Classifications
DNA	88,203,702	217,040	0.08	335
LINE	95,553,328	199,835	0.09	511
LTR	39,476,928	87,021	0.04	362
Other (Simple Repeat, Microsatellite, RNA)	37,933,124	77,043	0.04	112
Penelope	25,187,297	61,172	0.02	112
Rolling Circle	6,897,773	17,411	0.01	45
SINE	828,944	2039	0.00	6
Unclassified	404,513,462	966,499	0.39	1520

Table 4: Enrichment of annotated features among HCEs

Feature	Mean Enrichment	Median Enrichment	StDev Enrichment	N Chromosomes
3' UTR	0.99	1.02	0.26	16
5' UTR	0.81	0.78	0.37	16
CDS	1.94	1.98	0.85	16
TE	0.79	0.81	0.09	16
intron	1.42	1.23	0.67	16
lncRNA	0.95	0.91	0.35	16
multiple	1.34	1.35	0.15	16
other	0.90	0.88	0.11	16
rRNA	0.00	0.00	0.00	6
tRNA	4.99	4.10	3.36	16